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ORIGINAL ARTICLE

Low-load resistance training promotes muscular adaptation regardless of vascular occlusion, load, or volume

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Abstract
Purpose This study investigates the impact of two different intensities and different volumes of low-load resistance training (LLRT) with and without blood flow restriction on the adaptation of muscle strength and size.
Methods The sample was divided into five groups: one set of 20 % of one repetition maximum (1RM), three sets of 20 % of 1RM, one set of 50 % of 1RM, three sets of 50 % of 1RM, or control. LLRT was performed with (OC) or without (NOC) vascular occlusion, which was selected randomly for each subject. The maximal muscle strength (leg extension; 1RM) and the cross-sectional area (quadriceps; CSA) were assessed at baseline and after 8 weeks of LLRT.
Results 1RM performance was increased in both groups after 8 weeks of training: OC (1 × 50 % = 20.6 %; 3 × 50 % = 20.9 %; 1 × 20 % = 26.6 %; 3 × 20 % = 21.6 %) and NOC (1 × 50 % = 18.6 %; 3 × 50 % = 26.8 %; 1 × 20 % = 18.5 %; 3 × 20 % = 21.6 %; 3 × 20 % = 24.7 %) compared with the control group (−1.7 %). Additionally, the CSA was increased in both groups: OC (1 × 50 % = 2.4 %; 3 × 50 % = 3.8 %; 1 × 20 % = 4.6 %; 3 × 20 % = 4.8 %) and NOC (1 × 50 % = 2.4 %; 3 × 50 % = 1.5 %; 1 × 20 % = 4.3 %; 3 × 20 % = 3.8 %) compared with the control group (−0.7 %). There were no significant differences between the OC and NOC groups.
Conclusion We conclude that 8 weeks of LLRT until failure in novice young lifters, regardless of occlusion, load or volume, produces similar magnitudes of muscular hypertrophy and strength.

Keywords Strength · Cross-sectional area · Ischemia · Muscle mass · Hypertrophy

Abbreviations

1RM	One repetition maximum
95 % CI	95 % confidence intervals
CSA	Cross-sectional area
ES	Effect size
LLRT	Low-load resistance training
OC	LLRT with blood flow restriction
NOC	LLRT without blood flow restriction performed until volitional fatigue
MRI	Magnetic resonance imaging
NOC	Non-occlusion
OC	Occlusion
RT	Resistance training
SCSA	Six cross-sectional area images summed

Introduction

It is assumed that high-load (i.e., ≥70 % of one repetition maximum) and low repetition maximum (i.e., ≤12

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repetitions maximum) resistance training (RT) are necessary to induce significant increases in muscle strength and size (ACSM 2009). However, several studies have supported the efficacy of low-load RT (15–50 % of one repetition maximum) in promoting increased muscle mass and strength similar to those observed after high-load resistance training (Burd et al. 2010b, 2012; Mitchell et al. 2012; Loenneke et al. 2012; Abe et al. 2012). These adaptations in skeletal muscle from low-load resistance training (LLRT) are achieved when LLRT is performed either until volitional fatigue (i.e., failure) (Burd et al. 2010b, 2012; Mitchell et al. 2012) or under conditions of restricted blood flow (Loenneke et al. 2012; Abe et al. 2012). These findings are important because new interventions would be possible when high-load resistance training cannot be used (Manini and Clark 2009; Wernbom et al. 2008).

Understanding the relationship between the acute variables of training (load and amount of exercise) and skeletal muscle adaptation is important for creating efficient protocols (ACSM 2009; Martin-Hernandez et al. 2013; Loenneke et al. 2012). However, little is known regarding the effects of acute variables of LLRT on muscle mass and strength. LLRT with blood flow restriction (OC) has employed a training load ranging from 15 to 50 % of one repetition maximum (1RM) (Loenneke et al. 2012; Takarada et al. 2002). Interestingly, a dose–response relationship has been noted between OC load and muscle hypertrophy (Abe et al. 2012). In contrast, LLRT without occlusion performed until volitional fatigue (NOC) has typically employed 30 % of 1RM and has been found to promote hypertrophy similar to that of high-load resistance training (80–90 % of 1RM) (Burd et al. 2010b, 2012; Mitchell et al. 2012).

It has been suggested that these two conditions, OC and NOC, increase fiber recruitment to maintain muscle tension, and presumably, to stimulate muscle protein synthesis during recovery similarly to that of high-load resistance exercise (Loenneke et al. 2011; (Burd et al. 2010b, 2012; Mitchell et al. 2012). Additionally, Wernbom et al. (2013) found similar responses in hypertrophic signaling after a single episode of low-load resistance exercise performed to failure (30 % of 1RM) with and without blood flow restriction. Thus, it would seem reasonable to assume that performing repetitions until, or close to, volitional fatigue is an important stimulus for promoting muscle hypertrophy in both exercise conditions, regardless of load, or blood flow restriction.

High training volume (i.e., ≥3 sets) has been employed in LLRT (Loenneke et al. 2012; Abe et al. 2012; (Burd et al. 2010b, 2012; Mitchell et al. 2012). However, exercise until, or close to, volitional fatigue is not submaximal and therefore a high training volume may not be appropriate when high-load resistance training cannot be used (Scott et al. 2014). Recently, Martín-Hernández et al. (2013) found that increasing the OC volume from four (75 repetitions; volume = ~2133) to eight sets (150 repetitions; volume = ~4140) did not promote a further benefit to muscle size or strength. This study provides evidence for a possible volume threshold in which additional volume provides no further benefits. However, the absence of a volume below the four sets of OC in the study by Martín-Hernández et al. makes it impossible to identify the lowest volume threshold to induce adaptations of muscle size and strength. It has been evidenced that a single set (high-load) protocol stimulates myofibrillar protein synthesis (Burd et al. 2010a), hypertrophy, and muscular strength (ACSM 2009). However, there is no available information concerning a lower threshold below the three sets of LI-BRF or NOC that could be enough to induce significant muscular adaptations (Martin-Hernandez et al. 2013).

Acknowledging that the volitional fatigue is an important stimulus for promoting muscular adaptation, we hypothesized that similar gains in muscle size and strength could be detected regardless of occlusion or load when LLRT is performed until volitional fatigue. Moreover, we also hypothesized that a single set (very low volume) could be sufficient to accrete skeletal muscle mass and strength when LLRT is performed until volitional fatigue. To assess our hypotheses, we investigated the impact of two different loads and different volumes of LLRT performed until, or close to, volitional fatigue with and without blood flow restriction on muscle strength and size adaptations.

Materials and methods

Subjects

Forty-seven young men who were between the ages of 18 and 30 years and apparently healthy participated in this study (convenience sample). None of the subjects practiced periodic physical activity, had experience with RT, used anabolic steroids or nutritional supplements, had alcoholism, smoked, or used stimulants or medications that could affect muscle metabolism, and all subjects were free of risk factors for vascular disease. Subjects were chosen through a clinical history, completed before the start of the study, that was composed of an interview, a questionnaire that contained questions to identify indicators of peripheral vascular disease (personal and family history), and a visual inspection of the lower limbs of the individual. Furthermore, to verify diet homogeneity between the groups at the beginning of the study, subjects were instructed by a nutritionist to describe the foods eaten for 3 days, composed of 1 weekend day, and 2 week-days. The amounts of energy and macronutrients for each individual were obtained through the nutritional analysis software (DietPro Version 5i, Viçosa, MG, and Brazil).

All subjects were clear on the objectives and procedures for the study and gave their written informed consent. The study (No. 1984) was approved by the University Review Board for the Use of Human Subjects (local Ethics Committee) and was written in accordance with the standards set by the Declaration of Helsinki.

Experimental procedure

An experimental study was performed for 8 weeks. Initially, all subjects completed an evaluation of the cross-sectional area (CSA) of the quadriceps muscle by magnetic resonance imaging (MRI), body composition by anthropometric evaluation, and maximal muscle strength by a one repetition maximum test (1RM, see the 1RM section). All evaluations were repeated at the end of the eighth weeks of RT. After the initial evaluation, individuals were randomly (raffle) allocated to one of five conditions: (1) 1 × 20 % of 1RM, (2) 3 × 20 % of 1RM, (3) 1 × 50 % of 1RM, (4) 3 × 50 % of 1RM, and (5) control (CT). The subjects were then balanced within groups for leg dominance and randomly (raffle) selected by pairs (left or right) for these conditions: RT with vascular occlusion (OC) or RT without vascular occlusion (NOC). Subjects performed a unilateral knee extension exercise for 8 weeks. To avoid any residual influence of RT or vascular occlusion, the individuals trained each leg on separate days. Each leg was trained on two nonconsecutive days of the week between Monday and Friday with a minimum interval of 48 h (i.e., Monday—OC leg, Tuesday—NOC leg, Wednesday—rest, Thursday—OC leg, and Friday—NOC leg). To avoid differences in load or volume, RT was first performed on the occluded leg until concentric failure (inability to maintain range of motion), and then the same load and number of repetitions were applied to the non-occluded leg. This procedure was performed for all training sessions. The 1 × 20 % group performed one set of 20 % of one repetition maximum (1RM). The 3 × 20 % group performed a higher volume (three sets) than the 1 × 20 % group, but maintained the same relative load (20 % of 1RM). The 1 × 50 % group performed one set, but trained with a higher load (50 % of 1RM) than either the 1 × 20 or 3 × 20 % groups. The 3 × 50 % group performed a higher volume (three sets) than the 1 × 50 %, but maintained the same load (50 % of 1RM). All groups were allowed 60 s of rest between sets and up to one second for each muscle action (concentric and eccentric). The control group was instructed to perform only habitual physical activities and to avoid physical exercise or sports for 8 weeks.

Vascular occlusion was performed with a manual pneumatic tourniquet (ITS-MC, 28–100, Novo Hamburgo, SC, Brazil). The manual pneumatic cuff (80 cm of length and 10 cm of width) was placed around the proximal portion of the thigh. Initially, the cuff was inflated to 120 mmHg for the first training session. In the second and third training sessions, the cuff was increased by 30 mmHg each (180 mmHg). In the fourth training session, it was increased by 20 mmHg, reaching the maximum pressure of 200 mmHg. This pressure was used for all individuals for the remaining sessions of this study, (Madarame et al. 2008; Takarada et al. 2002, 2004). Vascular occlusion was maintained throughout the exercise and released at the end of the exercise. To avoid any differences in occlusion time between groups, the one set groups (1 × 20 and 1 × 50 %) remained occluded after the exercise for the same amount of time (approximately 5 min) as the groups that performed three sets (3 × 20 and 3 × 50 %). After the vascular occlusion progression, a drop of blood was collected from the subject's finger to analyze the lactate level immediately following all conditions and protocols.

Magnetic resonance imaging

In the morning hours (between 7:00 am and 9:00 am) and after an overnight fast of 8 h, subjects remained in the supine position for 1 h before the exam to avoid any influence of fluid shifts. Additionally, only regular physical activity, not strenuous exercise or activity, was allowed 48 h prior to the evaluation. Subjects underwent leg MRI performed with a 1.5 tesla unit (MAGNETOM Avanto: Siemens Healthcare, Erlangen, Germany), and cross-sectional images of both thighs were obtained before and after the intervention period. The magnetic field frequency was 65 MHz, the field of view was 372, and the transverse section had a thickness of 7.7 mm. The repetition and echo times of these sections were 5000 and 119 ms, respectively. Sections were obtained through coronal plane images, and an initial view of the lower limbs was made to determine the distance between the top line of the femoral head and the patellar face at an angle of 0° to each individual. This image served as a reference for the measurement of cross-sections of the thighs of the volunteers. Ten cross-sectional images were obtained of the thighs, with the first two and the last two discarded, for the analysis of the quadriceps muscle area (total: six images). These four images were discarded because they did not provide visualization of all quadriceps muscles. The obtained images were transferred to a computer for calculating the anatomical cross-sectional area using a specific ImageJ software plug-in for scanning. For an overall evaluation of the quadriceps muscle, the six cross-sectional area images of the leg were summed and used for comparisons (SCSA). Pre- and post-scans were performed at the same time of day, and joint angle and leg compression were controlled.

Table 1 Age, body mass index, body fat and perceptual energy, and macronutrient intake for all groups at baseline

	CT ($n = 8$)	1 × 50 % ($n = 10$)	3 × 50 % ($n = 10$)	1 × 20 % ($n = 10$)	3 × 20 % ($n = 10$)	$P^{\#}$
Body fat (%)	20.0 (9.3–26.1)	16.5 (11.5–23.2)	15.5 (7.01–18.1)	12.3 (8.3–25.0)	19.0 (13.0–26.3)	0.477
BMI (kg/m^2)	25.5 (21.3–30.0)	25.1 (22.2–29.3)	23.3 (21.3–25.5)	22.385 (20.9–26.4)	25.0 (21.5–29.0)	0.448
Age (years)	21.0 (20.0–27.2)	22.0 (20.0–24.0)	21.0 (19.1–25.0)	22.0 (19.9–23.5)	21.0 (20.0–24.0)	0.798
Energy (kcal)	1822.0 (1599.2–2106.3)	1828.5 (1473.5–2291.0)	2039.1 (1664.3–2968.0)	1775.0 (1702.2–2002.0)	1792.3 (1180.1–2350.4)	0.874
Fat (g/kg)	1.0 (0.5–1.1)	1.0 (1.0–1.1)	1.1 (1.0–1.3)	1.0 (1.0–1.0)	1.0 (0.5–1.2)	0.312
Protein (g/kg)	1.1 (1.0–1.5)	1.0 (1.0–1.4)	1.1 (1.0–2.0)	1.2 (1.0–1.3)	1.2 (1.0–1.3)	0.795
CHO (g/kg)	3.5 (2.0–4.5)	4.0 (3.0–4.3)	3.4 (2.1–5.5)	3.4 (2.5–4.0)	3.2 (2.0–5.4)	0.957

Median and 95 % CI

CT control group, *1 × 50* one set of 50 % of 1RM, *3 × 50 %* three sets of 50 % of 1RM, *1 × 20 %* one set of 20 % of 1RM, *3 × 20 %* three sets of 20 % of 1RM, *CHO* carbohydrate, *kcal* kilocalorie, *BMI* body mass index

Kruskal–Wallis test

To estimate reproducibility, eight study subjects (16 legs) had the six magnetic resonance images obtained on two consecutive days. Using the sum of the six images, a difference of 0.2 cm^2 (95 % CI −2.6 a 3.2 cm^2) or 0.1 % (95 % CI −0.6 a 0.8 %) was observed between the two evaluations. The correlation coefficient was 0.996, with a precision of 0.996 and an accuracy of 0.999.

1RM testing

Prior to the test phase, all subjects participated in three sessions, on alternating days, to become familiar with the exercise equipment and technique. The evaluation of maximal muscle strength was performed at three times during the study: test and retest pre-training (intraclass correlation coefficient = 0.98; 95 % CI 0.97–0.99) and again after 8 weeks of training. The 1RM test was performed for the unilateral knee extension exercise. The 1RM test began with the individual sitting with hips and knees at a 90° angle. Initially, a warm-up was performed using a subjective load, determined during the familiarization, with approximately 10 repetitions of 40–60 % of 1RM. After 1 min of rest, the load was increased, and three to five repetitions were performed with a subjective load of 60–80 % of 1RM. After 3 min of rest, the load was considerably increased, and the subjects were encouraged to overcome resistance using full motion. When the load was overestimated or underestimated, the subjects rested three to 5 min before a new attempt was performed with a lower or higher load, respectively. This procedure was performed to find the equivalent load of 1RM, which ranged between three and five tries. The load that was adopted as the maximum load was the load used for the last execution of the exercise that was performed with no more than one repetition by the subject.

Statistical analysis

The data were tested for normal distribution using the Shapiro–Wilk test and for variance homogeneity using the Levene test. The change (delta or delta %) in results of pre- and post-intervention was used for data comparison. The Kruskal–Wallis test was used for the comparison between groups. When appropriate, a post hoc comparison test of subgroups was made. Effects sizes were measured by Choen's r (nonparametric data; $r = \frac{Z}{\sqrt{N}}$) to compare pre- and post-values. Cohen's effect sizes (r) were interpreted as follows: $r < 0.1$ = null effect, $r < 0.3$ = small effect, $r < 0.5$ = medium effect, and $r \geq 0.5$ = large effect (Fritz et al. 2012). Data are presented as median values and 95 % confidence intervals (95 % CI). The level of significance was set at alpha ≤0.05.

Results

At baseline, no significant differences were observed between groups for age, BMI, body fat, macronutrients, energy, muscle strength (1RM), and cross-sectional area (Tables 1, 2).

There was no significant difference between training groups in the acute lactate response (Table 3). All groups performed different repetitions and volumes, but there was no significant difference between condition groups (OC and NOC) (Table 3). The total number of repetitions performed for each group, in order from lowest to highest, was as follows: 1 × 50, 3 × 50, 1 × 20, and 3 × 20 %. The 1 × 20 % group performed the lowest volume followed by 1 × 50 and 3 × 20 % groups, which performed the same volume. The 3 × 50 % group performed the highest volume.

Following the 8 weeks of training, 1RM performance was increased for all training groups compared with the

Table 2 Leg extension 1RM and quadriceps SCSA values for all groups at baseline (PRE) and post training (POST)

	CT ($n = 16$ legs)	OC 1 × 50 % ($n = 10$ legs)	NOC 1 × 50 % ($n = 10$ legs)	OC 3 × 50 % ($n = 10$ legs)	NOC 3 × 50 % ($n = 10$ legs)	OC 1 × 20 % ($n = 10$ legs)	NOC 1 × 20 % ($n = 10$ legs)	OC 3 × 20 % ($n = 10$ legs)	NOC 3 × 20 % ($n = 10$ legs)	$P^{\#}$
Leg extension 1RM										
PRE (kg)	122.0 (95.0–153.7)	81.0 (71.0–110.3)	83.5 (71.0–115.0)	95.0 (66.3–128.7)	101.0 (69.8–119.3)	84.5 (65.8–95.0)	85.0 (67.4–94.1)	75.0 (56.0–88.9)	73.0 (55.0–98.6)	0.100
POST (kg)	117.5 (96.1–149.2)	105.0 (83.3–137.9)	109.0 (91.9–138.8)	121.0 (92.5–152.9)	121.0 (94.0–145.6)	104.0 (85.0–113.6)	92.0 (79.4–113.6)	95.0 (72.1–119.6)	91.0 (72.4–122.2)	
Delta (kg)	−4.5 (−14.4 to 2.4)	16.5 (12.4–27.6)*	17.0 (9.0–32.1)*	28.1 (12.4–35.5)*,†	23.5 (15.2–36.8)*,†	19.0 (5.9–24.1)*	12.5 (5.8–20.3)*	15.0 (2.4–39.3)*	16.5 (2.9 − 30.1)*	<0.001
Effect size (r)	−0.20	0.63	0.63	0.63	0.44	0.63	0.58	0.59	0.63	
Quadriceps SCSA										
PRE (cm^2)	438.2 (385.5–501.7)	323.2 (367.4–443.4)	423.0 (358.7–454.1)	395.6 (376.7–444.1)	404.8 (381.5–440.6)	389.3 (341.8–438.9)	387.0 (336.5–438.3)	387.3 (364.0–439.5)	382.5 (355.6–438.3)	0.592
POST (cm^2)	432.4 (381.9–494.3)	439.6 (376.0–450.9)	431.6 (376.4–459.8)	414.8 (392.3–446.0)	413.2 (388.8–434.9)	407.1 (347.7–465.9)	405.8 (340.3–459.5)	412.1 (381.6–468.7)	401.5 (361.1–455.3)	
Delta (cm^2)	−3.3 (−6.2 to 0.1)	9.2 (4.7–19.3)*	10.1 (5.2–14.7)*	16.9 (3.6–18.9)*	5.8 (−1.0–23.5)*,‡	19.4 (2.9–30.6)*	18.8 (3.8–27.0)*	17.8 (8.9–31.6)*	16.9 (1.5–25.0)*	<0.001
Effect size (r)	−0.43	0.63	0.63	0.60	0.38	0.54	0.56	0.58	0.54	

Median and 95 % CI

CT control group, *1 × 50* one set of 50 % of 1RM, *3 × 50 %* three sets of 50 % of 1RM, *1 × 20 %* one set of 20 % of 1RM, *3 × 20 %* three sets of 20 % of 1RM, *SCSA* sum of quadriceps cross-sectional areas, *1RM* one repetition maximum, *OC* RT with vascular occlusion, *NOC* RT without vascular occlusion

Kruskal–Wallis test

* Difference ($P < 0.05$) from CT group

† Difference ($P < 0.05$) from NOC 1 × 20 % group

‡ Difference ($P < 0.05$) from OC 3 × 20 % group

Table 3 Lactate, leg extension repetitions, and volume for all groups

	OC 1 × 50 % (*n* = 10 legs)	NOC 1 × 50 % (*n* = 10 legs)	OC 3 × 50 % (*n* = 10 legs)	NOC 3 × 50 % (*n* = 10 legs)	OC 1 × 20 % (*n* = 10 legs)	NOC 1 × 20 % (*n* = 10 legs)	OC 3 × 20 % (*n* = 10 legs)	NOC 3 × 20 % (*n* = 10 legs)	*P*[#]
Lactate (mMol/L)	5.6 (5.0–7.1)	5.6 (5.0–7.0)	6.2 (6.0–8.0)	6.0 (5.0–7.2)	6.5 (5.5–8.0)	6.0 (5.0–7.2)	7.0 (6.0–9.0)	6.0 (4.4–9.0)	0.376
Repetitions per day	25 (21–30)[a]	25 (21–29)[a]	42 (35–58)[b]	42 (37–58)[b]	51 (42–58)[b]	50 (43–58)[b]	76 (54–90)[c]	76 (63–90)[c]	<0.001
Volume load per day (kg) (product of load in kg and repetitions completed)	1112.5 (824.3–1757.1)[a]	1191.7 (786.8–1854.9)[a]	2509.5 (1480.1–3346.6)[b]	2383.3 (1602.3–3276.2)[b]	891.4 (732.2–1042.6)[c]	865.9 (732.7–983.3)[c]	1397.9 (725.4–2328.2)[a]	1590.2 (710.9–1793.6)[a]	<0.001

Median and 95 % CI

Values are presented as median and 95 % CI

SCSA sum of quadriceps cross-sectional areas, *CT*, control group, *1 × 50* one set of 50 % of 1RM, *3 × 50 %* three sets of 50 % of 1RM, *1 × 20 %* one set of 20 % of 1RM, *3 × 20 %* three sets of 20 % of 1RM, *OC* RT with vascular occlusion, *NOC* RT without vascular occlusion

[#] Kruskal–Wallis test

Different letters indicate significant differences between groups ($P < 0.05$)

Identical letters indicate no differences between groups

control group. There was significant difference between NOC-1 × 20 and 3 × 50 % (OC and NOC) in the 1RM absolute change (Table 2). However, there were no significant differences between the training groups for the 1RM % change (Fig. 1). Furthermore, Cohen's *r* for muscle strength indicates a large effect size for all training groups except for NOC-3 × 50 % which showed medium effect (Table 2).

Similarly, the sum of quadriceps cross-sectional areas (SCSA) was increased in all training groups compared with the control group. There was a significant difference between NOC-3 × 50 and OC-1 × 20 % for the cross-sectional area absolute change (Table 2). However, there were no significant differences between training groups for the cross-sectional area % change (Fig. 2). Furthermore, Cohen's r for muscle strength indicates a large effect size for all training groups except for NOC-3 × 50 % which showed medium effect (Table 2).

There were no harms to subjects.

Discussion

To our knowledge, this is the first study comparing the effects of LLRT with one or three sets of different intensities performed until failure, with and without blood flow restriction on muscle adaptation in strength and size. The main finding of the present study was that there was no significant difference in the magnitude of quadriceps muscle hypertrophy (as determined by MRI) or muscular strength between the doses of LLRT performed until failure, with or without vascular occlusion, after 8 weeks of knee-extensor exercise.

In situations where high-load resistance training cannot be used, new interventions to increase muscle mass and strength are needed. There is an evidence supporting the effects of OC and NOC on increasing muscle mass and strength (Burd et al. 2010b; Loenneke et al. 2012; Mitchell et al. 2012). However, little is known regarding the most effective protocol to improve performance. The data from the present study indicate that there is no significant difference between OC and NOC for increasing SCSA and leg extension 1RM strength. It has been reported that high muscle fiber recruitment and the activation of type II fibers are necessary to induce hypertrophy (ACSM 2009). Although during a single repetition more motor units are recruited with increasing requirement for force generation (Henneman's size principle), when a submaximal contraction is sustained to failure, the recruitment of additional motor units is necessary to sustain muscle tension (Burd et al. 2012; Fallentin et al. 1993). In this context, studies have proposed that repetitions performed close to or until volitional fatigue with a light load result in additional

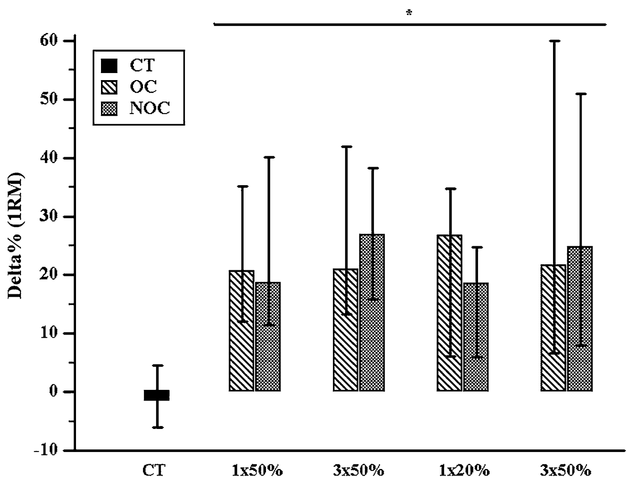

Fig. 1 Perceptual changes (delta) of leg extension 1RM for all groups and conditions. Values are presented as median values and 95 % CIs. *SCSA* sum of quadriceps cross-sectional areas, *CT* control group, *1 × 50* one set of 50 % of 1RM, *3 × 50 %* three sets of 50 % of 1RM, *1 × 20 %* one set of 20 % of 1RM, *3 × 20 %* three sets of 20 % of 1RM, *OC* RT with vascular occlusion, *NOC* RT without vascular occlusion. *Significantly different from the CT group ($P < 0.05$)

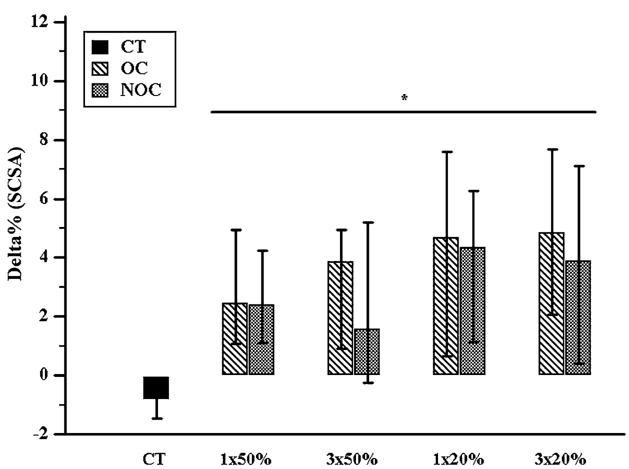

Fig. 2 Perceptual changes (delta) of quadriceps SCSA for all groups and conditions. Values are presented as median values and 95 % CIs. *SCSA* sum of quadriceps cross-sectional areas, *CT* control group, *1 × 50* one set of 50 % of 1RM, *3 × 50 %* three sets of 50 % of 1RM, *1 × 20 %* one set of 20 % of 1RM, *3 × 20 %* three sets of 20 % of 1RM, *OC* RT with vascular occlusion, *NOC* RT without vascular occlusion. *Significantly different from the CT group ($P < 0.05$)

motor units to sustain muscle tension, thereby promoting an important stimulus for muscle hypertrophy and strength (Burd et al. 2012; Loenneke et al. 2011; Burd et al. 2010b; Mitchell et al. 2012). Indeed, studies utilizing biopsies and fibers metabolite analysis have demonstrated that type II fibers may be recruited after 1.5-min bout (i.e., 90 repetitions) of knee-extensor exercise at a target frequency of 60 extension/min with an external power output of ~65 W (110 % of peak thigh VO2) and without occlusion, which would have exhausted the subjects after about ~4-min (240 repetitions) (Krustrup et al. 2004b, 2009). Furthermore, all fibers may be recruited during a 3-min bout (180 repetitions) of the same knee-extensor exercise protocol, (Krustrup et al. 2004b) and there is a homogeneous recruitment of the quadriceps muscle portions at high exercise intensity (Krustrup et al. 2009). However, when the intensity of exercise is reduced to 50 % of peak thigh VO2 solely, type I fibers are recruited and there is no homogeneous recruitment of the quadriceps muscle portions (Krustrup et al. 2004a, 2009). Therefore, a homogeneous recruitment of the quadriceps muscle portions and additional fibers are recruited with time to sustain muscle tension during intense, but not moderate, submaximal exercise (Krustrup et al. 2004a, 2009). However, when the load of the same knee-extensor exercise protocol (i.e., 60 extension/min) is reduced by ~55 % (i.e., ~29 W or 50 % of peak thigh VO_2), but the same exercise time is kept (i.e., 1.5-min or 90 repetition), type II fibers are recruited in the occlusion condition but not without occlusion condition (Krustrup et al. 2009). These findings suggest that there is a significant afferent response related to the metabolic stress from the contracting muscles affecting the activation of fibers in the contracting muscles (Krustrup et al. 2004a, b, 2009). Although these previously mentioned studies are not OC studies specifically, they do provide some insight to the nature of occlusion and fatigue per se on muscle fibers recruitment patterns. Based on the repetition number in the previously mentioned studies, the necessary load percentage to achieve an intense knee-extensor exercise seems to be below the lowest load used in our study. For instance, the volunteers in the present study performed ~50 repetition maximum (i.e., 1.6-min bout) at 20 % of 1RM, while the studies cited above (which performed knee-extensor exercise at 110 % of peak thigh VO_2) achieved failure at ~240 repetitions (i.e., 4-min bout of the knee-extensor exercise at a target frequency of 60 extension/min), suggesting that they used an even lighter load to achieve an intense knee-extensor exercise (Krustrup et al. 2004b, 2009). Moreover, the lactate response observed in our study suggests that the knee-extensor exercise was intense (Krustrup et al. 2004a) and a similar effort for both legs, regardless of occlusion. Using an experimental design similar to ours, Wernbom et al. (2013) recently showed that acute LLRT with and without occlusion increased protein signaling and the number of satellite cells in human skeletal muscle. Therefore, LLRT (≥20 % of 1RM) performed close to or until volitional fatigue results in a similar amount of muscle fiber recruitment and a homogeneous recruitment of muscle portions, promoting similar muscular adaptation with or without vascular occlusion.

Traditionally, it is accept that neural adaptations increase strength during the early stages of training and muscle hypertrophy becomes evident within the first 6 week (ACSM 2009). Our findings are in concordance with this adaptation. We found that the strength increased by 22.3 % (CI 18.6–26.3), whereas the muscle mass increased by only 3.6 % (CI 2.2–4.3) for all training groups after 8 weeks. This result showed that the relative strength (i.e., the maximal strength per unit of muscle size) of muscles trained was changed from pre-training levels, suggesting that neural adaptations occurred early. In contrast, Loenneke et al. (2012) found a significant correlation between strength development and weeks of OC but not for muscle hypertrophy, suggesting that neural adaptations occur later and that initial increases in strength may be due solely to muscle hypertrophy. However, this meta-analysis failed to find studies that used low load close to concentric failure, which may explain the difference between the studies. Although our and previous studies indicate that LLRT is capable of producing increased strength in previously untrained individuals, a longer intervention period (>8 weeks) could have revealed significant differences in 1RM between the intensities, volumes, or occlusion conditions (NOC and OC) because it has been suggested that neural adaptations for increased strength may occur later with OC training (Loenneke et al. 2012). However, several studies have showed beneficial muscular adaptations (strength, power, and agility) to OC training in athletes (Cook et al. 2014; Manimmanakorn et al. 2013a, b; Yamanaka et al. 2012). Thus, future research is needed to address this issue.

It has been suggested that there is a relationship between load and hypertrophy, (ACSM 2009) or muscular strength (ACSM 2009; Mitchell et al. 2012) for RT without occlusion but only between load and muscle hypertrophy for RT with occlusion (Abe et al. 2012). However, our results indicate that there is no difference between 20 and 50 % of 1RM for increases in SCSA and leg extension 1RM strength after LLRT performed until failure regardless of blood flow restriction. Although there were differences between NOC-1 × 20 % and OC/NOC-3 × 50 % for the 1RM absolute change and between NOC-3 × 50 and OC-1 × 20 % for the SCSA absolute change, there were no significant differences between training groups for the relative change (% delta). Furthermore, Cohen's *r* for muscle strength or SCSA indicates a large effect size for all training groups except for NOC-3 × 50 % which showed medium effect (Table 2). Despite a medium effect for NOC-3 × 50 %, the probability of superiority for an *r* of 0.38 is ~72 %. That is, if we sampled items randomly, one from each moment (pre- and post-intervention), the one from the post intervention would be higher than the one from the pre intervention for 72 % of the comparisons. The probability of superiority for a *r* of 0.50 (large effect) is ~80 % (Fritz et al. 2012). Thus, our results suggest that performing low-load (<50 % of 1RM) knee extensions close to concentric failure promotes similar muscular adaptation (strength and hypertrophy) regardless of load (20 vs. 50 %). As discussed previously, LLRT completed to the point of failure results in a similar muscular adaptation. Indeed, Mitchell et al. (2012) showed that NOC (30 % of 1RM) resulted in similar hypertrophy as a heavy RT (80 % of 1RM) completed to the point of failure. This study showed that NOC significantly increases the unilateral knee extension 1RM after 10 week. Our results are also consistent with those observed in other studies in which OC has been shown to induce gains in muscle CSA and 1RM strength to a similar extent as that of high-load RT (Kubo et al. 2006; Karabulut et al. 2010; Laurentino et al. 2012).

Although a high number of sets and repetitions have been used in LLRT (Loenneke et al. 2012; Burd et al. 2010b, 2012; Mitchell et al. 2012; Abe et al. 2012; Martin-Hernandez et al. 2013; Wernbom et al. 2013), Martín-Hernández et al. (2013) recently showed that doubling the OC volume from four to eight sets resulted in no further benefit in muscle size or strength. The possibility of a volume threshold has been suggested in which additional volume provides no further benefits for both traditional resistance training (Gonzalez-Badillo et al. 2006) and OC (Martin-Hernandez et al. 2013). However, information is lacking for the chronic effects of LLRT with very low volume and for the chronic effects of volume of LLRT performed until failure on muscle adaptation. Our study results indicate that initial gains in both SCSA and leg extension 1RM strength are not affected by the volume of LLRT performed until failure. We found gains in both SCSA and leg extension 1RM strength using LLRT performed until failure even with very low volume (~866) and repetitions (~25). Thus, a very low threshold volume (1 × 20 %, ~866 kg) or number of repetitions (1 × 50 %, ~25) of LLRT performed until failure is sufficient to induce significant increases in muscle mass and strength in previously untrained individuals after 8 weeks.

One possible limitation of the present study is the cross-education effect. The unilateral training utilized in this study can lead to neural adaptation and strength gains in the contra-lateral leg (Lee and Carroll 2007). However, this limitation would only affect the conditions within groups and not between groups (different subjects). Because there was no significant difference between groups for hypertrophy or strength gains, we can support our results regardless of this limitation. Furthermore, Mitchell et al. (2012) found no correlation between legs (left and right) for strength gains after different RT protocols, suggesting that the cross-education effect is minimal when both limbs are trained.

In conclusion, a similar magnitude of muscle hypertrophy and muscular strength can be achieved from 8 weeks of low-load resistance training (≤50 % of 1RM) performed until failure regardless of blood flow restriction, load or volume in young, novice lifters.

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Conflict of interest The authors declare that they have no conflict of interest.

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